BraTS MRI Reconstruction Project

Generated from config `config/formal_train.yaml` on 2026-04-28 04:50:12.

Group Information

- Course: BME AI Project 1
- Group: 第 4 组
- Chinese names: 唐志昊 2022533131
- Note: Metadata has been filled automatically. Review the final report before submission.

AI Usage Declaration

This project used AI tools for code debugging, experiment automation, and report drafting. Manual review and result verification were performed before submission.

Presentation reminder: Remember to claim AI usage in the presentation slides as required by the assignment.

Project Objective

This project reconstructs fully sampled T2 brain MRI slices from AF=5 undersampled k-space using the BraTS dataset. The assignment requires three tasks: undersampling simulation, a baseline reconstruction model, and a multi-modal unrolled model with data consistency.

Dataset and Preprocessing

- Dataset path: `../bmeaidataset`
- Modalities used: T1, T2
- Slice axis: 2
- Intensity normalization: z-score on non-zero voxels

- Split strategy: patient-level train/validation/test
- Slice selection for this run: 16 central slices per patient
- Volume loading mode: preloaded in RAM to reduce I/O stalls
- Split counts: train=6992, validation=1504, test=1504

Methods

Task 1

A 2D random variable-density sampling mask with acceleration factor 5 is generated in k-space. Fully sampled T2 slices are transformed with FFT, masked, and reconstructed with inverse FFT to obtain aliased images.

Task 2

Task 2 uses a U-Net baseline with base channels 24, depth 4, batch size 8, MSE loss, and learning rate 0.0005. `ReduceLROnPlateau` is used for learning rate decay.
During training, `channels_last`, pinned memory, non-blocking GPU transfers, TF32, and multi-worker prefetching are enabled to reduce GPU idle time.

Task 3

Task 3 uses an unrolled U-Net with 2 cascades, data consistency layers, and multi-modal input consisting of aliased T2 plus fully sampled T1. The loss is `hybrid` with L1 weight 0.7.

Division of Labor

- 唐志昊：完成项目整合、训练实验执行、结果核查、报告整理与提交。

Quantitative Results

- Task 2 improved PSNR from 25.76 dB to 35.18 dB and SSIM from 0.3628 to 0.9271.
- Task 3 improved PSNR from 25.76 dB to 38.12 dB and SSIM from 0.3628 to 0.9660.
- Compared with Task 2, Task 3 changed PSNR by 2.93 dB and SSIM by 0.0389.

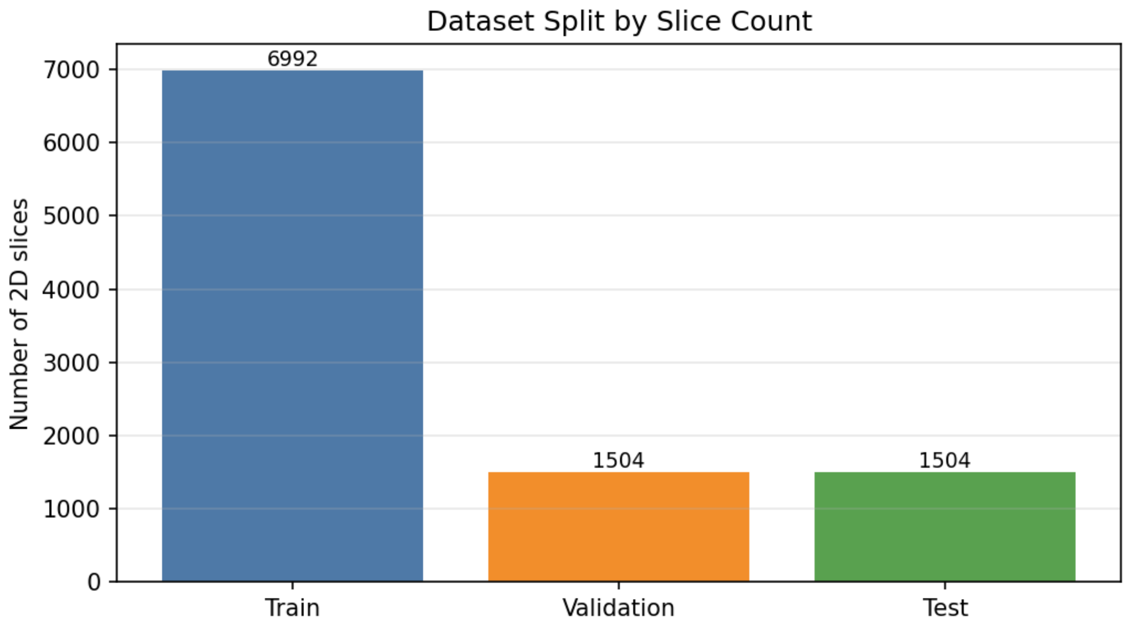

Figure 1: Dataset Split Chart

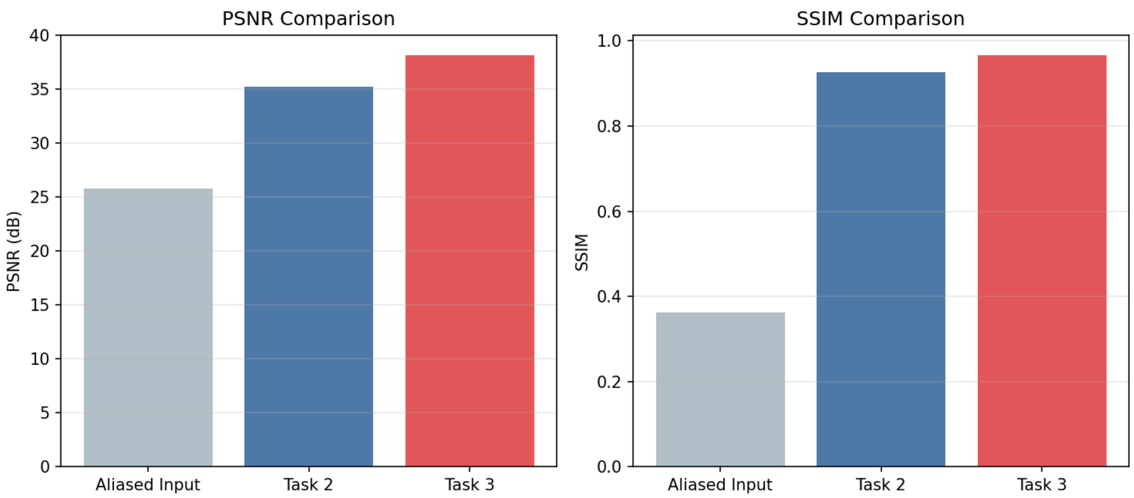

Figure 2: Metric Comparison Chart

Figures and Tables

Dataset Split Chart

Metric Comparison Chart

Summary Tables

Dataset Split

Split	Slice Count
Train	6992
Validation	1504
Test	1504

Hyperparameters

Setting	Task 2	Task 3
Input	Aliased T2	Aliased T2 + full T1
Backbone	U-Net	Unrolled U-Net + DC
Base channels	24	16
Depth	4	3
Batch size	8	4
Epochs	10	8
Learning rate	0.0005	0.0001
Loss	MSE	hybrid

Quantitative Results

Method	PSNR (dB)	SSIM
Aliased input	25.76	0.3628
Task 2 baseline	35.18	0.9271
Task 3 multi-modal	38.12	0.9660

Improvements Over Aliased Input

Method	PSNR Gain (dB)	SSIM Gain
Task 2 baseline	9.43	0.5643
Task 3 multi-modal	12.36	0.6032

Task 3 Over Task 2

Comparison	Value
PSNR gain	2.93 dB
SSIM gain	0.0389

Worst-case Table

Rank	Patient	Slice	PSNR Before	PSNR After	SSIM Before	SSIM After
1	BraTS-GLI-00077-000	76	23.26	34.69	0.2865	0.9443
2	BraTS-GLI-00077-000	82	23.14	34.74	0.2875	0.9502
3	BraTS-GLI-00088-001	82	24.33	34.82	0.3160	0.9439
4	BraTS-GLI-00077-000	78	23.23	34.82	0.2942	0.9481
5	BraTS-GLI-00077-000	79	23.22	34.84	0.2921	0.9484

Task 1 Visualization

Task 2 Loss Curve

Task 2 Reconstructions

Task 3 Loss Curve

Task 3 Best Reconstructions

Task 3 Error Analysis

Error Analysis

The 5 worst-performing Task 3 cases are summarized below.

Rank	Patient	Slice	PSNR After	SSIM After
1	BraTS-GLI-00077-000	76	34.69	0.9443
2	BraTS-GLI-00077-000	82	34.74	0.9502
3	BraTS-GLI-00088-001	82	34.82	0.9439
4	BraTS-GLI-00077-000	78	34.82	0.9481
5	BraTS-GLI-00077-000	79	34.84	0.9484

Potential improvements for these cases include stronger edge-preserving loss terms, more cascades if runtime allows, and targeted inspection of slices with complex tumor boundaries.

Execution Status

- Current run status: Success
- Output directory: `outputs_formal`
- Total runtime: 24m 42s

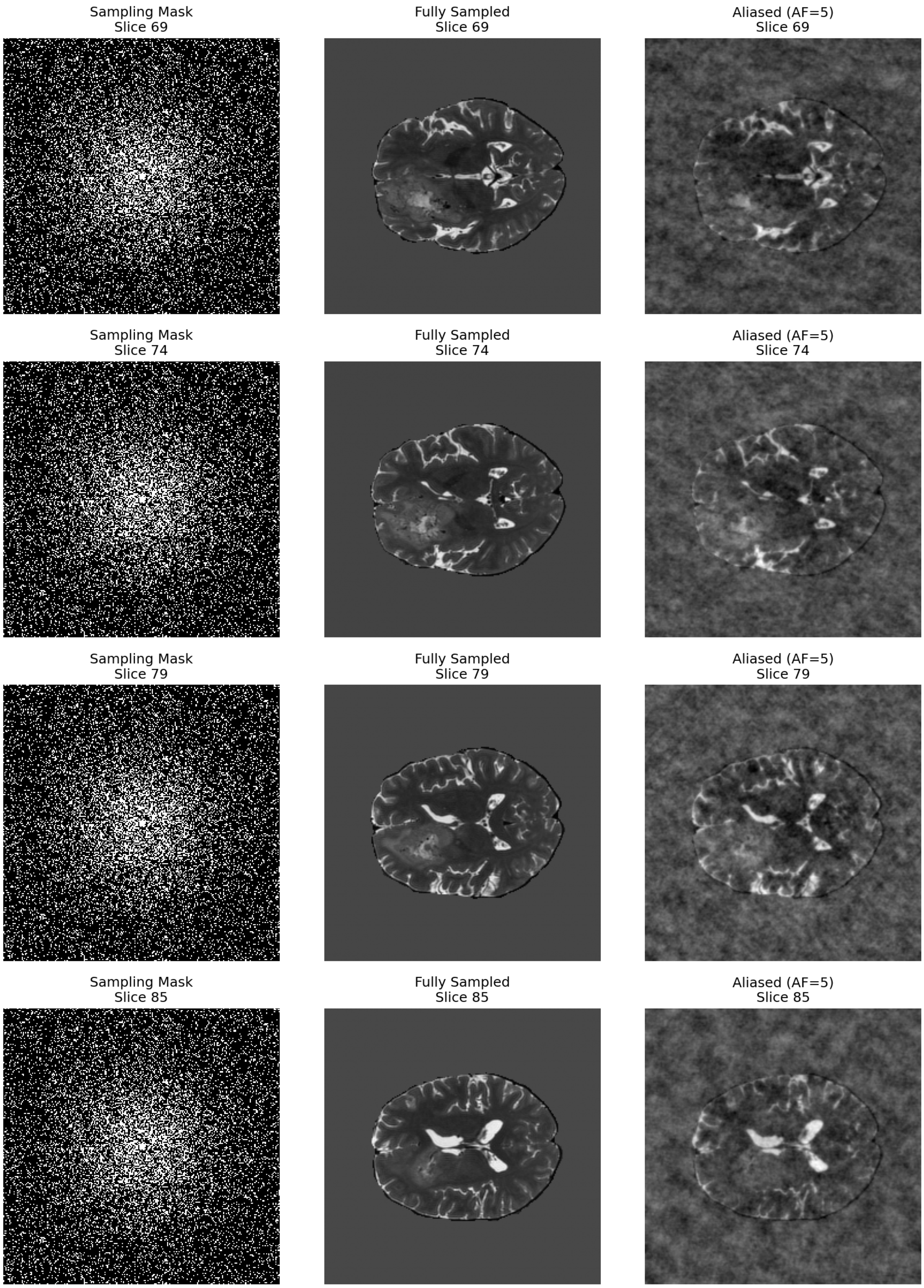

Figure 3: Task 1 Visualization

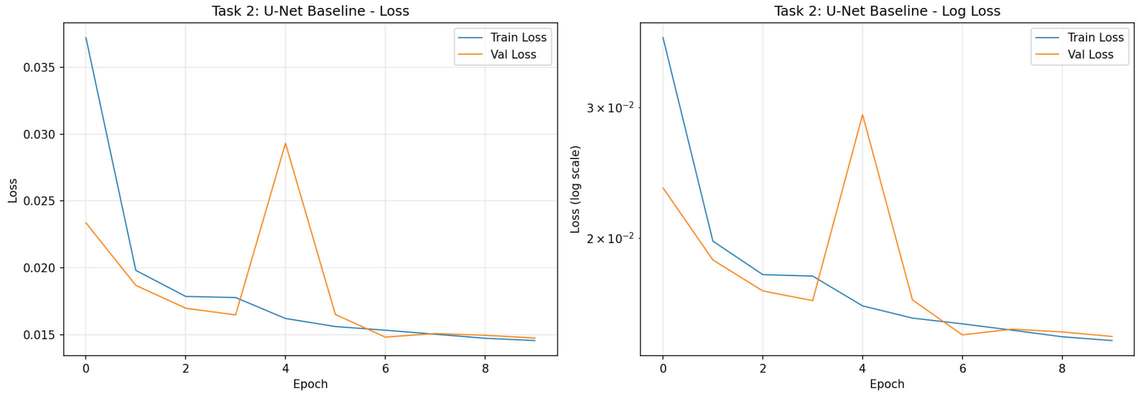

Figure 4: Task 2 Loss Curve

- Last completed stage: assets
- Log file: `outputs_formal/logs/formal_pipeline.log`

Discussion

The current results support the expected conclusion of the project: the baseline network substantially reduces aliasing artifacts, and the multi-modal unrolled model provides additional measurable improvement while using 16 central slices per patient across all available patients.

Submission Checklist

- Code: prepared
- Report draft: `REPORT.md`
- LaTeX report: `REPORT.tex`
- PDF report: `REPORT.pdf`
- Presentation slides: still need to be prepared manually

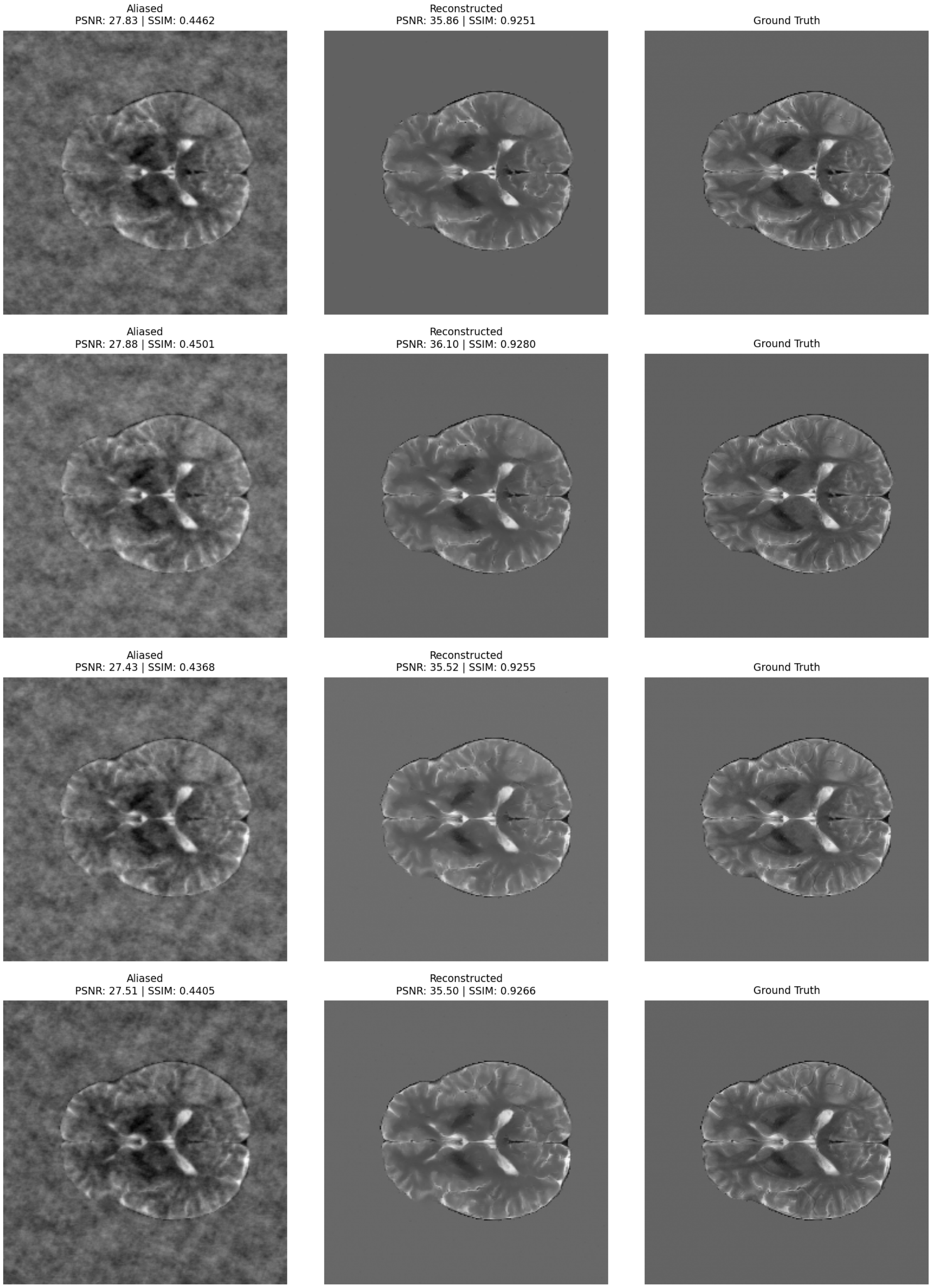

Figure 5: Task 2 Reconstructions

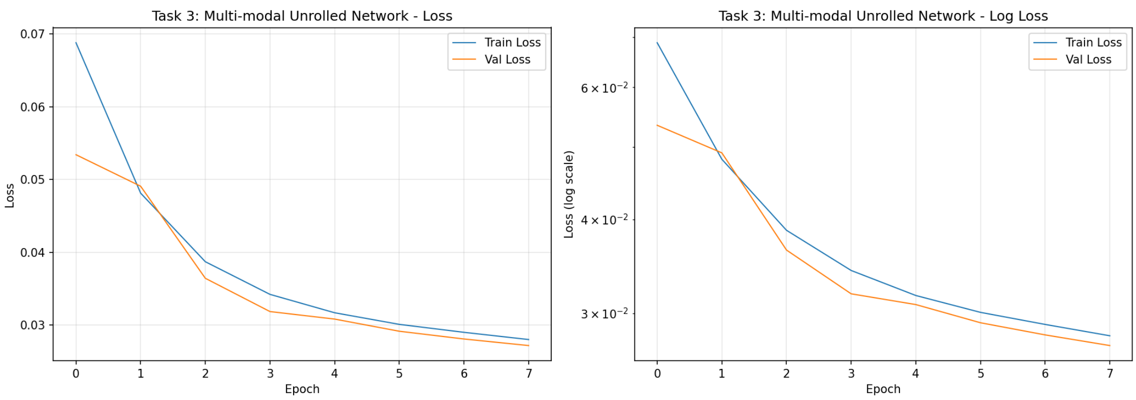

Figure 6: Task 3 Loss Curve

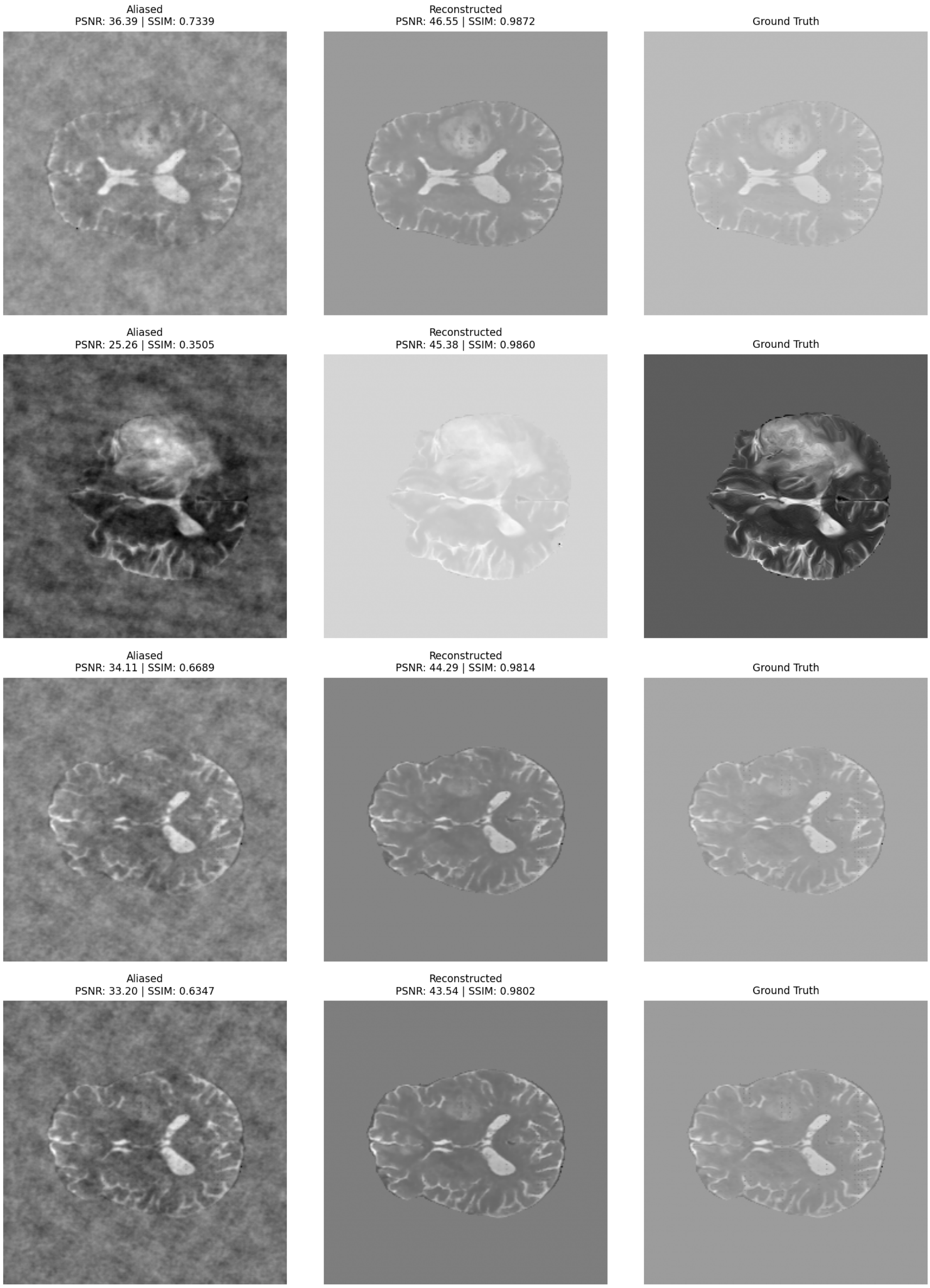

Figure 7: Task 3 Best Reconstructions

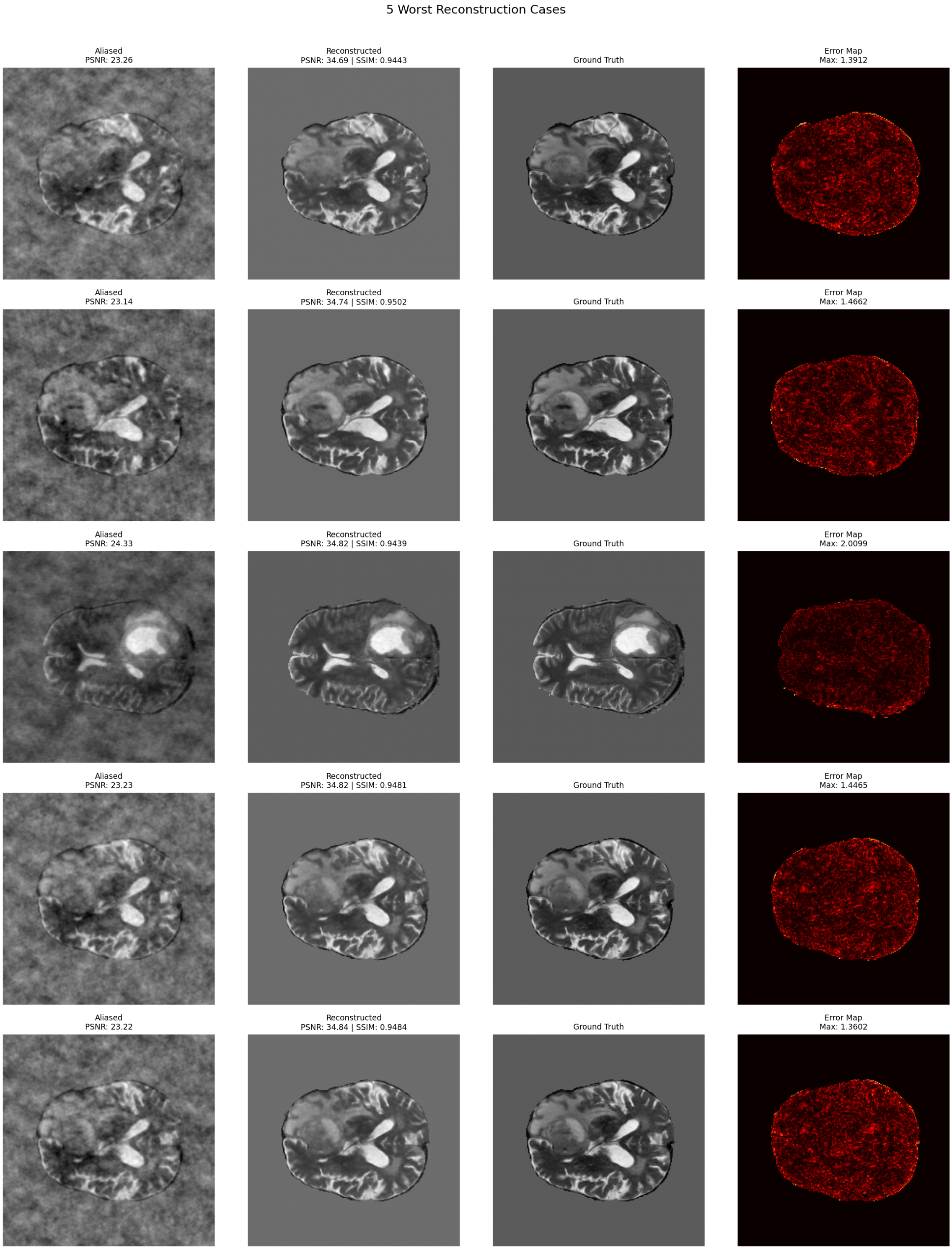

Figure 8: Task 3 Error Analysis